

# Data Warehouses and Data Mining Systems

Laboratory 3: Report content

## 1 Data warehouse datasheet

The data warehouse datasheet is a form of organizing information describing the future data warehouse. It should contain information about the facts, dimensions etc. as entities, relationships and attributes, as well as some additional information about the variability in time, data source, technical details etc.

## 2 Logical model

The logical model of the data warehouse should be developed, basing on the conceptual model defined previously. The logical model is in fact an ERD diagram – it contains entities and relationships (1:N, N:M) between them. The logical model should contain the following elements:

1. Fact(s) – containing all defined measures (foreign keys are not required, since it is not yet a physical model);
2. Dimensions - containing all defined attributes and existing hierarchies;
3. Relationships between facts and dimensions – usually 1:N (when there are N:M relationships, a *bridge* pattern should be applied).

The logical model is usually organized as a star or snowflake schema, where the fact is placed in the center of the diagram, while being surrounded by all dimensions.